

# PBSMS

## Frontend Design Specification

Version 1.1 · Companion to SRS v2.1, Chapter 34

Complete interface proposal across all system modules

| Item                | Value                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Document            | PBSMS Frontend Design Specification                                                                                                |
| Version             | 1.1 — audited against SRS v2.1 Ch. 34 & 42, gaps closed                                                                            |
| Status              | For review. No part of this has been user-tested; see §15.                                                                         |
| Governs             | SRS v2.1 Chapter 34 (Frontend Architecture, UX & Offline/PWA Strategy) and Chapter 42 (Accessibility)                              |
| Scope               | All modules across SRS Volumes 0–V, expressed as four frontend surfaces                                                            |
| Companion prototype | pbsms-frontend-prototype.html, manifest.json, sw.js — interactive, PWA-capable when served (not opened as a local file — see §9.2) |

### Revision Note — v1.0 → v1.1

v1.0 was audited against SRS v2.1 Chapter 34 and Chapter 42 line by line. Six requirements were not met as written, and one design decision deviated from the SRS's stated pattern without being flagged as a deviation. This revision closes all seven. Nothing in this list is a new feature — every item was already a stated requirement in v1.0's own source document; v1.0 simply didn't satisfy it yet.

| # | Requirement                                  | Was                                                                                      | Now                                                                                                                                                            |
|---|----------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 34.1 — active campus visible                 | Context switcher showed School only                                                      | Switcher shows School and Campus (§6.1)                                                                                                                        |
| 2 | 34.1 — breadcrumbs, responsive drawer        | Named in the component library, never specified                                          | Breadcrumb pattern and a dedicated Staff Console small-screen spec (§6.1.1)                                                                                    |
| 3 | FR-UX-020 — offline score entry              | Only attendance was detailed; scores were a one-line inventory entry                     | Full detailed spec, same rigor as the register (§8.11)                                                                                                         |
| 4 | FR-UX-010 / FR-UX-040 — PWA, data-saver      | Asserted in prose, not designed; prototype had no manifest or service worker             | Manifest, service-worker caching strategy, and a concrete data-saver mode specified (§9.2, §9.3); companion manifest.json and sw.js shipped with the prototype |
| 5 | 34.5 — six required UI states                | Three of six (loading, empty, restricted) had no spec or example                         | All six specified with examples (§8.12)                                                                                                                        |
| 6 | NFR-ACC-020 — CI-enforced accessibility gate | Written in this document; absent from the actual CI pipeline                             | Added to .github/workflows/ci.yml in the code scaffold (§11.2)                                                                                                 |
| 7 | 34.5 — hide vs. disable unavailable actions  | Platform Console disabled blocked actions instead of hiding them, presented as compliant | Named explicitly as an approved, reasoned deviation (§14) rather than left unflagged                                                                           |

# 1. Purpose & Relationship to the SRS

SRS v2.1 Chapter 34 specifies frontend behaviour — an application shell, a design-token layer, offline capability for attendance and score entry, WCAG 2.1 AA, and distinct UI states. It does not specify a single screen. This document fills that gap and nothing else: it proposes the interface, and it defers to the SRS on every functional question.

Where this document and the SRS disagree, the SRS wins. Where this document proposes something the SRS does not cover, it is flagged as a proposal, not a requirement.

## 1.1 What this document contains

- A design-token system with real values (§4) — Chapter 34.2 requires tokens but defines none.
- Four frontend surfaces (§3), because 'the frontend' is not one product.
- A complete screen inventory across every module in SRS Volumes 0–V (§7).
- Detailed specifications for the highest-risk screens (§8).
- Offline and sync UX, satisfying FR-UX-010..040, with PWA infrastructure and data-saver mode (§9).
- Role-based visibility rules mapped to Chapter 13 scope (§10).
- Accessibility implementation detail against WCAG 2.1 AA (§11).
- A performance budget tied to NFR-PERF-020 (§12).
- Every deviation from the SRS's literal text, named and justified rather than left implicit (§14).

## 2. Design Principles

These are derived from constraints in the SRS and in the market, not from general design preference. Each one is falsifiable — you can tell whether a screen obeys it.

| Principle                         | Derived from                                                                                        | What it forbids                                                                                                             |
|-----------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| The register is the product       | Attendance and score entry are the highest-frequency actions in the system (FR-ATT-010, FR-ASM-020) | Burying score entry three levels deep behind an academics menu. A teacher reaches either in one tap from their home screen. |
| Never lie about sync              | FR-UX-030                                                                                           | Optimistic checkmarks. A row is 'saved on this phone' or 'saved to the school' and the interface always says which.         |
| Money is auditable on screen      | Chapter 23.1 — every balance derived from valid transactions                                        | A balance shown as a single number with no path to the transactions that produced it.                                       |
| Published is different from draft | FR-RES-030 — published results are never overwritten                                                | Any visual treatment where a draft result and a published result look alike.                                                |
| Data before decoration            | NFR-PERF-020 (2.5s on 3G), FR-UX-040 (low data)                                                     | Hero images, icon fonts, decorative illustration, webfont stacks over 100KB.                                                |
| One tenant at a time              | TEN-012, TEN-013                                                                                    | Any screen that mixes two tenants' data, including in platform support tooling.                                             |

### 3. The Four Surfaces

The single most consequential proposal in this document: PBSMS is not one interface. It is four, sharing one design system, one component library and one codebase. Designing them as one app produces a desktop admin tool that teachers cannot use on a phone in a classroom — which is the failure mode that kills school software adoption.

| Surface           | Primary user                                                                   | Device reality                                                                 | Design posture                                                                                    |
|-------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Staff Console     | Head, administrator, accountant, coordinator, exam officer, admissions officer | Laptop or shared desktop, office, reliable power                               | Data-dense. Tables, filters, bulk actions, keyboard shortcuts. Desktop-first, responsive down.    |
| Teacher Field App | Class and subject teachers                                                     | Personal Android phone, in a classroom, connectivity unreliable, one hand free | Offline-first. One task per screen, large touch targets, thumb-zone actions, works fully offline. |
| Parent View       | Guardians                                                                      | Low-cost Android, metered data, often arrives from a WhatsApp link             | Read-mostly, extremely light. No app install required. Loads under 2.5s on 3G or it has failed.   |
| Platform Console  | PBSMS staff (Chapter 3 roles)                                                  | Laptop, internal                                                               | Deliberately austere. Tenant-scoped, audit-visible, no bulk destructive actions.                  |

These share: tokens, components, auth, and the offline/sync layer. They do not share navigation, information density, or layout. A parent never sees a sidebar; a teacher never sees a data table.

## 4. Design Tokens

Chapter 34.2 requires 'central design tokens' governing colour, typography, spacing and breakpoints, with a per-tenant branding layer on top. Here they are. Every value below is a proposal open to change — but the system needs concrete values, and 'we will decide later' is how design debt starts.

### 4.1 Colour — rationale

The palette comes from the subject's own materials: the deep green-black of a classroom blackboard, the warm manila of an exercise book, and the gold that already signals attainment in Ghanaian schools and on the national flag. It is deliberately not a generic SaaS blue.

Gold is reserved. It marks attainment and nothing else — a passing grade, a completed term, a settled account. If gold starts appearing on buttons and headers it stops meaning anything, and the one visual signal a parent scanning a report card actually needs is lost.

#### Core

| Token          | Value   | Use                                          | Contrast on canvas |
|----------------|---------|----------------------------------------------|--------------------|
| --pb-ink       | #12211F | Body text, headings                          | 14.8:1 — AAA       |
| --pb-ink-muted | #5A6663 | Secondary text, table meta, captions         | 5.4:1 — AA         |
| --pb-teal-900  | #0B4F4A | Primary actions, headers, active nav         | 9.1:1 — AAA        |
| --pb-teal-600  | #12726B | Hover, links, focus ring                     | 5.9:1 — AA         |
| --pb-teal-50   | #E6EFEC | Selected rows, subtle fills                  | —                  |
| --pb-gold      | #9A6700 | Attainment only — grades, settled, completed | 5.2:1 — AA         |
| --pb-gold-50   | #FBF3E0 | Attainment fills, award chips                | —                  |
| --pb-canvas    | #FAF7F0 | Page background (warm, paper)                | —                  |
| --pb-surface   | #FFFFFF | Cards, tables, sheets                        | —                  |
| --pb-rule      | #E3DED2 | Hairlines, table borders, dividers           | —                  |

Every contrast ratio in this table is calculated, not tool-verified. Re-check each one against a contrast checker before implementation — see §11.2 and §15.

#### Semantic

| Token        | Value   | Meaning in this product                             |
|--------------|---------|-----------------------------------------------------|
| --pb-success | #1E6B3A | Synced, posted, approved, present                   |
| --pb-warning | #8A5A00 | Pending review, overdue soon, unsynced              |
| --pb-danger  | #A32017 | Failed, reversed, absent, overdue, validation error |
| --pb-info    | #12726B | Neutral system state                                |

Colour is never the only carrier of meaning (WCAG 1.4.1). Every state above pairs with a shape, an icon, or text — see §11.1.

### 4.2 Typography

Two faces, one of them free. The body face is the device's own system stack, which costs zero bytes and renders familiarly on cheap Android — a direct concession to FR-UX-040. One display face is loaded, subset, for headings and figures.

| Role               | Face                                                             | Why                                                                                                                                                                                 |
|--------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Display / headings | Archivo (variable, subset to Latin, woff2, ~28KB)                | A grotesque with enough width and weight range to carry dense admin headers without a second face. Not a high-contrast serif — those read as editorial, and this is a working tool. |
| Body / UI          | system-ui stack — Roboto on Android, SF on iOS, Segoe on Windows | Zero network cost, instantly familiar, no FOUT on the screens teachers use most.                                                                                                    |

| Role                  | Face                                                                         | Why                                                                                                                                                                                                                 |
|-----------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Figures & identifiers | Archivo with font-variant-numeric: tabular-nums; identifiers in ui-monospace | Scores, money and attendance counts must align in columns. Admission numbers, invoice numbers and BECE index numbers are read aloud and cross-checked against paper — monospace makes transcription errors visible. |

### Type scale (1.200 minor third, 16px base)

| Token          | Size | Weight | Use                                         |
|----------------|------|--------|---------------------------------------------|
| --pb-text-2xs  | 11px | 500    | Table meta, timestamps                      |
| --pb-text-xs   | 12px | 500    | Labels, chips, captions                     |
| --pb-text-sm   | 14px | 400    | Table cells, secondary body                 |
| --pb-text-base | 16px | 400    | Body. Never smaller for parent-facing text. |
| --pb-text-lg   | 19px | 500    | Card titles, section heads                  |
| --pb-text-xl   | 23px | 600    | Page titles                                 |
| --pb-text-2xl  | 28px | 600    | Dashboard figures                           |
| --pb-text-3xl  | 33px | 700    | Teacher app primary figures, parent balance |

## 4.3 Space, radius, elevation, motion

| Group            | Tokens                                                                                                                                                                                        |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Space (4px base) | --pb-space-1 4px · -2 8px · -3 12px · -4 16px · -5 24px · -6 32px · -7 48px · -8 64px                                                                                                         |
| Radius           | --pb-radius-sm 4px (inputs, chips) · -md 8px (cards, sheets) · -lg 12px (modals) · -full (avatars only)                                                                                       |
| Elevation        | --pb-shadow-1 0 1px 2px rgba(18,33,31,.08) · -2 0 4px 12px rgba(18,33,31,.10) · -3 0 12px 32px rgba(18,33,31,.14). Three levels only; anything needing a fourth is a layout problem.          |
| Motion           | --pb-ease standard cubic-bezier(.2,0,0,1) · --pb-dur-fast 120ms (state) · -base 200ms (sheets, drawers) · -slow 320ms (page transitions). All motion suppressed under prefers-reduced-motion. |
| Breakpoints      | --pb-bp-sm 480px · -md 768px · -lg 1024px · -xl 1280px. Teacher app and Parent view are designed at 360px first — the real floor for Android in this market.                                  |

## 4.4 Tenant branding layer

Per Volume 0 Chapter 6, branding is per tenant. Exactly three things are tenant-overridable, resolved once at the layout root and never per screen:

- --pb-brand-primary — replaces teal in nav, primary buttons and document letterheads. Validated on upload for 4.5:1 contrast against white; rejected with a clear message if it fails, because a tenant must not be able to configure their own school into an inaccessible interface.
- Logo — used in the shell header, report cards, receipts and certificates.
- School display name and letterhead block.

Everything else — type, spacing, semantic colour, component shape — is fixed. Tenants get identity, not a theme engine. This keeps the component library testable and stops a school from breaking its own accessibility conformance.

## 5. Component Library

Built once, used across all four surfaces. Naming follows what users control, not how it is built (a person manages notifications, not webhook config).

| Group        | Components                                                                                                                                                                                                      |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Foundational | Button (primary/secondary/ghost/danger), IconButton, Link, Chip, Badge, Avatar, Tag, Divider, Spinner, ProgressBar                                                                                              |
| Form         | TextField, NumberField (tabular figures), Select, Combobox, DatePicker (term-aware), Checkbox, Radio, Toggle, Textarea, FileUpload, ScoreInput (bounded, keyboard-fast), CurrencyInput (GHS-aware), SearchField |
| Data         | DataTable (sortable, filterable, paginated, column-configurable), DefinitionList, KeyFigure, TrendSparkline, EmptyState, StatusPill, AuditTrail, Timeline, ComparisonBar                                        |
| Navigation   | AppShell, Sidebar, TopBar, ContextSwitcher (school/campus/year/term), Breadcrumbs, Tabs, Stepper, BottomNav (mobile), CommandPalette                                                                            |
| Feedback     | Toast, InlineAlert, ConfirmDialog, DestructiveConfirm (types-to-confirm), FieldError, FormSummaryError, OfflineBanner, SyncLedger                                                                               |
| Surface      | Card, Sheet (mobile bottom sheet), Modal, Drawer, Popover, Tooltip, Accordion                                                                                                                                   |
| Domain       | RegisterRow, ScoreGrid, ResultCard, GradeChip, InvoiceLine, PaymentAllocator, BalanceSummary, ReportCardPreview, CompetencyMeter, ApprovalBar, ConsentToggle                                                    |

The Domain group is where this product differs from any generic admin template. A ScoreGrid that understands assessment weighting, a PaymentAllocator that splits one payment across invoice lines, and a CompetencyMeter that renders NaCCA indicator coverage cannot be bought off the shelf and should be built with the most care.

## 6. Navigation & Information Architecture

### 6.1 Staff Console

Sidebar grouped by how school staff actually think about their work, not by SRS volume order. Groups collapse; items are permission-generated (Chapter 34.1).



The context switcher is now four-part — school, campus, academic year, term — not three. Chapter 34.1 requires the active campus to be visible, not only the active school; a tenant with one school and one campus can collapse the first two into a single control, but the underlying state (which campus is active) always exists and is always shown once a tenant has more than one, because that is exactly the case a single collapsed control would hide.

Fixed v1.0 defect: the switcher previously showed School only. Campus is Chapter 2's actual isolation-adjacent unit below School, and Volume 0 exists specifically to support multi-campus tenants — a shell that cannot show which campus is active does not meet the requirement it claims to.

#### Breadcrumbs

Required by 34.1, absent from v1.0. Breadcrumbs sit directly under the top bar on any screen more than one level deep, and are themselves navigable — clicking 'JHS 2A' from a student profile returns to that class's roster, not to a generic student list.



On the Teacher Field App and Parent View, breadcrumbs do not appear — those surfaces are shallow by design (§3), and a back arrow in the header is sufficient. Breadcrumbs are a Staff Console and Platform Console pattern only.

#### 6.1.1 Staff Console — small screens (fixes a v1.0 gap)

34.1 requires 'a responsive drawer on small screens.' v1.0 named this in the surfaces table as 'desktop-first, responsive down' and never specified what 'down' meant. It means the following, concretely:

| Breakpoint           | Sidebar              | Top bar                                                                   | Tables                      |
|----------------------|----------------------|---------------------------------------------------------------------------|-----------------------------|
| ≥1024px (--pb-bp-lg) | Persistent, expanded | Full — all four context controls, search, notifications, user menu inline | Full DataTable, all columns |

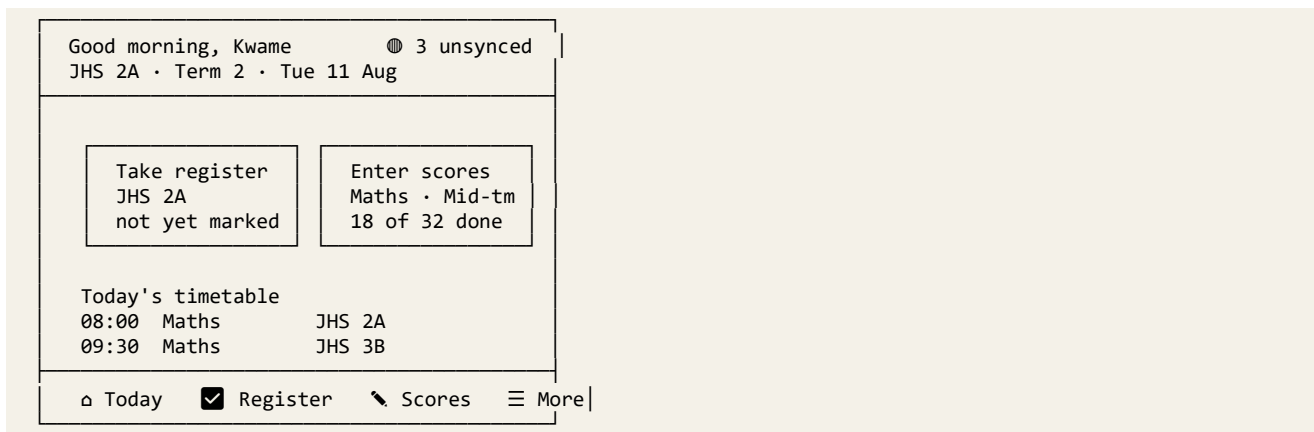


| Breakpoint              | Sidebar                                                                                                                     | Top bar                                                                                             | Tables                                                                                             |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 768–1023px (--pb-bp-md) | Collapses to icon rail; expands as an overlay on tap                                                                        | Context switcher collapses into a single 'School · Campus · Term' summary button that opens a sheet | DataTable drops secondary columns behind a 'more' control per row                                  |
| <768px (--pb-bp-sm)     | Hidden by default; opens as a full-height Drawer (component, §5) triggered by a menu icon, closes on selection or scrim tap | Reduced to: menu icon, breadcrumb (current screen only), user avatar                                | Tables convert to stacked cards — one record per card, primary fields only, tap through for detail |

This matters more than it looks: a proprietor checking collections from a phone between school visits is a real, common case for this product, not an edge case — Chapter 8's Proprietor role is explicitly the one most likely to be mobile.

## 6.2 Teacher Field App

No sidebar. Four bottom-nav destinations, sized for one-handed use, with the two highest-frequency tasks reachable in one tap from launch.



## 6.3 Parent View

No navigation chrome at all on first load. A parent arriving from a WhatsApp link sees the thing they were notified about, immediately, with everything else one tap below it. Multiple children appear as a horizontal switcher, not a menu.

## 6.4 Platform Console

Tenant list → tenant detail. Every screen states which tenant is in view, in a persistent bar that is visually distinct from the tenant-facing product (dark, unmistakable) so PBSMS staff can never confuse the console with a customer's own account. Impersonation adds a second, louder banner naming the ticket and the expiry time (TEN-021).

## 7. Screen Inventory — All Modules

Complete coverage of SRS Volumes 0–V. Surface key: S = Staff Console, T = Teacher Field App, P = Parent View, X = Platform Console.

### 7.1 Platform & Tenancy (Volume 0)

| Screen                                                   | Surface | SRS reference    |
|----------------------------------------------------------|---------|------------------|
| Tenant list (search, status filter, plan, student count) | X       | Ch 2, Ch 4       |
| Tenant detail — profile, schools, subscription, health   | X       | Ch 2, Ch 5       |
| Tenant lifecycle actions — suspend, offboard, reinstate  | X       | Ch 4.1           |
| Plan & pricing configuration                             | X       | Ch 5.1           |
| Metering & usage — active students per tenant per period | X       | TEN-030, TEN-031 |
| Platform invoices & dunning status                       | X       | FR-BIL-010..040  |
| Impersonation request & active-grant banner              | X       | TEN-021          |
| Platform audit log                                       | X       | TEN-022          |
| Platform revenue dashboard (MRR, churn, ARPU)            | X       | FR-BIL-050       |

### 7.2 Onboarding & Tenant Settings

| Screen                                                         | Surface | SRS reference |
|----------------------------------------------------------------|---------|---------------|
| Signup / trial start                                           | S       | Ch 4.1        |
| Onboarding wizard — school, year, terms, divisions, levels     | S       | FR-ONB-010    |
| Starter data review (grading scale, fee categories, templates) | S       | FR-ONB-010    |
| Bulk import — upload, map columns, dry-run validation, commit  | S       | FR-ONB-020    |
| Import error report & re-upload                                | S       | FR-ONB-020    |
| Branding — logo, primary colour, letterhead                    | S       | Ch 6, Ch 34.4 |
| Users & roles, delegation, conflict-of-interest warnings       | S       | Ch 13         |
| Integrations — payment, messaging, credential health           | S       | Ch 36         |
| Subscription & billing (tenant side)                           | S       | TEN-031       |

### 7.3 Dashboards (Chapter 14)

| Screen                                                          | Surface | SRS reference |
|-----------------------------------------------------------------|---------|---------------|
| Proprietor dashboard — group roll-up across schools             | S       | FR-ANL-010    |
| Headmaster dashboard — approvals, attendance, collections       | S       | Ch 14.1       |
| Accountant dashboard — today's collections, unverified, overdue | S       | FR-FIN-040    |
| Teacher home — register + scores + timetable                    | T       | Ch 14.1       |
| Parent home — child summary, balance, latest results            | P       | FR-STU-060    |
| Student home — own results, timetable                           | P       | FR-RES-040    |
| KPI configuration                                               | S       | Ch 14.2       |
| Action tracker & escalations                                    | S       | FR-COM-060    |

### 7.4 Admissions (Chapter 15)

| Screen                                                       | Surface | SRS reference |
|--------------------------------------------------------------|---------|---------------|
| Application list — state filter, ageing, source              | S       | FR-ADM-010    |
| Application detail — identity, guardians, documents, medical | S       | FR-ADM-010    |

| Screen                                                  | Surface | SRS reference |
|---------------------------------------------------------|---------|---------------|
| Duplicate-match review (side-by-side, merge or dismiss) | S       | FR-ADM-020    |
| Interview / entrance assessment scheduling & outcome    | S       | FR-ADM-010    |
| Approval & conversion to student (atomic, confirm step) | S       | FR-ADM-030    |
| Admissions KPIs — volume, processing time, conversion   | S       | FR-ADM-050    |

## 7.5 Students & Guardians (Chapter 16)

| Screen                                                   | Surface | SRS reference |
|----------------------------------------------------------|---------|---------------|
| Student list — filter by class, status, division, campus | S       | FR-STU-010    |
| Student profile shell with tabs (below)                  | S       | FR-STU-010    |
| · Identity & enrolment history                           | S       | FR-STU-010    |
| · Guardians & relationship flags                         | S       | FR-STU-020    |
| · Academics — results by term and year                   | S       | FR-RES-050    |
| · Attendance — trend and exceptions                      | S       | FR-ATT-050    |
| · Finance — invoices, payments, balance                  | S       | Ch 24         |
| · Health (restricted access)                             | S       | FR-OPS-030    |
| · Discipline (restricted access)                         | S       | FR-OPS-040    |
| · Documents                                              | S       | FR-DOC-010    |
| · Timeline — every significant event, chronological      | S       | FR-STU-050    |
| Transfer — class, campus, inbound, outbound              | S       | FR-STU-030    |
| Guardian directory & linked children                     | S       | FR-STU-020    |

## 7.6 Academic Structure (Chapter 17)

| Screen                                                | Surface | SRS reference       |
|-------------------------------------------------------|---------|---------------------|
| Academic years & terms — states, windows              | S       | FR-ACA-020          |
| Divisions, class levels, classes                      | S       | FR-ACA-010          |
| Subjects by division/level (+ optional NaCCA mapping) | S       | FR-ACA-030, DOM-010 |
| Teacher assignment & workload view                    | S       | FR-ACA-050          |
| Timetable builder with live conflict detection        | S       | FR-ACA-040          |
| Timetable views — class, teacher, room                | S       | FR-ACA-040          |

## 7.7 Attendance (Chapter 18)

| Screen                                     | Surface | SRS reference |
|--------------------------------------------|---------|---------------|
| Daily register — offline-capable           | T       | FR-ATT-010    |
| Register (desktop fallback)                | S       | FR-ATT-020    |
| Correction with reason (retains original)  | S       | FR-ATT-030    |
| Class & school attendance summary          | S       | FR-ATT-050    |
| Absence follow-up queue & guardian contact | S       | FR-ATT-040    |
| Staff attendance                           | S       | Ch 18.1       |
| Child attendance view                      | P       | FR-STU-060    |

## 7.8 Assessment, Grading & Results (Chapters 19–21)

| Screen                                                     | Surface | SRS reference          |
|------------------------------------------------------------|---------|------------------------|
| Assessment configuration & weighting (must total 100%)     | S       | FR-ASM-010             |
| Score entry grid — desktop, keyboard-fast                  | S       | FR-ASM-020             |
| Score entry — offline-capable mobile                       | T       | FR-ASM-020, FR-UX-020  |
| Grading scale & policy versions                            | S       | FR-GRA-030, FR-GRA-070 |
| Grading run — preview, blocking errors, no silent defaults | S       | FR-GRA-060             |
| Results workflow queue (draft → published)                 | S       | FR-RES-010             |
| Result review & return-with-comment                        | S       | FR-RES-010             |
| Publication gate — checklist of unmet conditions           | S       | FR-RES-020             |
| Published results & version history                        | S       | FR-RES-030             |
| Reopen with reason & authority                             | S       | FR-RES-030             |
| Result analytics — trend, class, subject, division         | S       | FR-RES-050             |
| Child results view                                         | P       | FR-RES-040             |

## 7.9 Documents & Promotion (Chapter 22)

| Screen                                            | Surface | SRS reference            |
|---------------------------------------------------|---------|--------------------------|
| Report card batch generation with progress        | S       | FR-DOC-010, NFR-PERF-023 |
| Report card template editor & preview             | S       | FR-DOC-030               |
| Certificate issue (JHS completion)                | S       | FR-PRO-010               |
| Public document verification lookup (QR target)   | P       | FR-DOC-020               |
| Promotion run — system recommendation vs decision | S       | FR-PRO-010, FR-PRO-020   |
| Promotion approval & next-year enrolment creation | S       | FR-PRO-030               |

## 7.10 Finance (Chapters 23–25)

| Screen                                               | Surface | SRS reference          |
|------------------------------------------------------|---------|------------------------|
| Fee structure builder — year, term, level, category  | S       | FR-FEE-010             |
| Instalment plan & proration rules                    | S       | FR-FEE-020, FR-FEE-030 |
| Penalty rules                                        | S       | FR-FEE-040             |
| Invoice generation run & preview                     | S       | FR-PAY-040             |
| Invoice list & detail with line items                | S       | Ch 24.1                |
| Payment capture — cash, MoMo, card, transfer         | S       | FR-PAY-010             |
| Payment allocation across invoice lines              | S       | Ch 24.1                |
| Receipt issue & reprint                              | S       | Ch 24.4                |
| Financial assistance — scholarship, discount, waiver | S       | FR-FIN-010             |
| Reversal request & approval (maker-checker)          | S       | FR-FIN-020             |
| Reconciliation workspace — provider vs internal      | S       | FR-FIN-030             |
| Student statement of account                         | S       | Ch 24.1                |
| Finance dashboard                                    | S       | FR-FIN-040             |
| Invoice, balance & pay now                           | P       | FR-PAY-010             |

## 7.11 Communication (Chapter 26)

| Screen                                      | Surface | SRS reference |
|---------------------------------------------|---------|---------------|
| Compose — channel picker with cost estimate | S       | FR-COM-040    |

| Screen                                                 | Surface | SRS reference |
|--------------------------------------------------------|---------|---------------|
| Template library & version editor                      | S       | FR-COM-020    |
| Audience builder — class, level, arrears, custom       | S       | FR-COM-040    |
| Send review — recipient count, channel split, cost     | S       | FR-COM-030    |
| Delivery log — per recipient, per channel, retry state | S       | FR-COM-040    |
| Consent registry & withdrawal handling                 | S       | DP-070        |
| Message thread                                         | P       | FR-COM-060    |

## 7.12 Analytics (Chapter 27)

| Screen                                                  | Surface | SRS reference |
|---------------------------------------------------------|---------|---------------|
| School analytics — attendance, performance, collections | S       | FR-ANL-020    |
| Group roll-up comparison across schools                 | S       | FR-ANL-030    |
| Cohort & trend explorer                                 | S       | FR-ANL-020    |
| AI-drafted summary with human-approval gate             | S       | FR-ANL-040    |

## 7.13 Supporting Operations (Chapter 28)

| Screen                                                | Surface | SRS reference |
|-------------------------------------------------------|---------|---------------|
| Library — catalogue, circulation, fines               | S       | FR-OPS-010    |
| Transport — routes, stops, vehicles, assignments      | S       | FR-OPS-020    |
| Health — records, incidents, medication log           | S       | FR-OPS-030    |
| Discipline — case, investigation, response, appeal    | S       | FR-OPS-040    |
| Inventory — assets, stock, issuance, low-stock alerts | S       | FR-OPS-050    |

## 7.14 Compliance & Curriculum (Volume V)

| Screen                                                    | Surface | SRS reference    |
|-----------------------------------------------------------|---------|------------------|
| Consent registry per guardian per channel                 | S       | DP-070           |
| Data subject request queue with 30-day SLA timer          | S       | DP-030, DP-090   |
| Retention & archive status by record type                 | S       | Ch 40.2          |
| Audit log viewer (tenant-visible, incl. platform actions) | S       | TEN-022          |
| NaCCA curriculum mapping — strand/indicator tree          | S       | DOM-010          |
| Competency coverage report                                | S       | DOM-020          |
| BECE candidate registration & index numbers               | S       | DOM-040          |
| Mock BECE results & readiness analytics                   | S       | DOM-050, DOM-060 |
| GES statutory return export                               | S       | DOM-070          |
| CSSPS choices & placement outcome                         | S       | DOM-080          |

## 8. Key Screen Specifications

Ten screens carry most of the product risk — they are the highest-frequency, the highest-consequence, or the hardest to retrofit. Specified here in detail; the rest follow the patterns they establish.

### 8.1 Daily Register (Teacher Field App) — the signature screen

Highest-frequency action in the entire system, performed in a classroom, often with no connectivity. Everything about this screen is subordinate to one goal: a teacher marks 32 children in under 45 seconds and knows exactly what has and has not reached the school.

|                                            |              |  |
|--------------------------------------------|--------------|--|
| ← JHS 2A · Tue 11 Aug                      | Mark all ▾   |  |
| 📶 Working offline · 12 marks on this phone | ← SyncLedger |  |
| Ama Mensah                                 | [P] A L E S  |  |
| ✓ on this phone                            |              |  |
| Kofi Boateng                               | P [A] L E S  |  |
| ✓ saved to school                          |              |  |
| Yaa Owusu                                  | P A [L] E S  |  |
| ✓ on this phone · reason required ⚠        |              |  |
| 29 of 32 marked                            |              |  |
| [ Submit register ]                        |              |  |

- Status is a five-state segmented control per row (Present, Absent, Late, Excused, Sick), not a dropdown — one tap, no modal, per FR-ATT-020.
- Each row states its own sync state in words, never a bare icon: 'on this phone' vs 'saved to school'. This is FR-UX-030 taken literally.
- The full roster is pre-fetched when the app last had connectivity, so the screen opens with data even in airplane mode.
- 'Submit register' works offline — it queues. The button never lies: offline it reads 'Submit when back online', and the SyncLedger count increments.
- Rows are 56px minimum, targets 44×44 minimum (WCAG 2.5.5), ordered by surname, sticky search for large classes.

### 8.2 SyncLedger (component, all Teacher Field App screens)

The proposed signature element of the product. Instead of a spinner, a persistent honest ledger of what has left the device.

📶 Working offline · 12 marks on this phone [View]  
 🔄 Sending · 4 of 12 saved to school  
 ✓ All 12 saved to school · 2 min ago  
 ⚠ 2 could not be saved · [Review]

- Always visible on register and score screens. Never a toast, because a toast disappears and the question 'did my marks save?' does not.
- Tapping it opens the queue: what is pending, what failed, what the failure was, and what to do about it.
- Conflicts (FR-ATT-011) surface here for manual reconciliation when two different users marked the same student — never silently resolved.
- Colour is paired with a distinct glyph and a text label, so the state is legible without colour vision.

### 8.3 Score Entry Grid (Staff Console)

A spreadsheet-grade editing experience, because that is what it replaces and users will compare it to Excel.

- Keyboard-first: Enter moves down, Tab moves right, values commit on blur, no save button per cell.
- Bounds enforced at the input (FR-ASM-030); an out-of-range value is rejected inline with the valid range stated, not silently clamped.
- Missing scores require an explicit status — absent, exempt, pending — never left ambiguous.

- Weighting bar shows live total; publication is blocked, with the reason stated, while weights do not sum correctly (FR-ASM-010).
- Locked after submission with a visible lock state and a 'request reopen' action rather than a disabled control with no explanation (FR-ASM-040).

## 8.4 Results Publication Gate (Staff Console)

FR-RES-020 lists five conditions for publication. The screen renders them as a live checklist, not an error after the fact.

Publish • JHS 2A • Term 2 • 2026/2027

- ✓ All 9 subjects approved
- ✓ Class teacher review complete
- ✓ Headmaster approval recorded
- ⚠ 2 students missing Social Studies scores  
Ama Mensah • Kofi Boateng [Resolve]
- ✓ Report template available

[ Publish ] – disabled, with the reason stated

A disabled button with no explanation is the most common failure in approval UI. Here the reason is always adjacent to the control, and every unmet condition links to the screen that resolves it.

## 8.5 Payment Allocation (Staff Console)

Chapter 24.1's model — payments independent of invoices, joined by allocations — is unusual and must be visible in the interface, or cashiers will misunderstand it.

- Left: the payment (amount, method, reference, provider status). Right: outstanding invoice lines.
- Allocation is explicit — drag or enter amounts per line, with a running unallocated remainder that must reach zero or be deliberately left as credit.
- Unallocated credit is a first-class visible state, not an error.
- Reversal is never a delete. The action reads 'Reverse payment', requires a reason, and produces a linked correcting entry shown in the same timeline (FR-FIN-020).

## 8.6 Parent Home (Parent View)

Assume arrival from a WhatsApp link on a metered connection. Under 2.5s on 3G, under 150KB total, no login wall for published content reached via an authenticated link.

|                                |                   |
|--------------------------------|-------------------|
| Sunrise Basic School           |                   |
| • Ama    ○ Kwesi               | ← child switch    |
| Term 2 results are ready       |                   |
| [ View report card ]           |                   |
| Balance                        |                   |
| GH¢ 450.00 due 30 Aug          |                   |
| [ Pay now ]                    | [ See statement ] |
| Attendance this term      96%  |                   |
| Present 44 • Absent 2 • Late 1 |                   |

Note what is absent: no sidebar, no dashboard grid, no KPI cards. A guardian has three questions — how is my child doing, what do I owe, and is anything wrong — and the screen answers them in that order.

## 8.7 Report Card (Parent View & print)

- One layout that renders on a 360px screen and prints to A4 without a separate template.
- Grade chips use gold for attainment, but the grade is always also stated in text.
- Where the tenant uses NaCCA (DOM-030), the competency profile appears above raw scores, because that is what the curriculum is actually assessing.
- The published version and date are always shown. If a result was later revised, both versions are reachable — FR-RES-030 makes this a data guarantee and the UI must not hide it.

## 8.8 Reconciliation Workspace (Staff Console)

- Two columns: provider settlement vs internal records, with matched, unmatched and disputed lanes.
- Provider fees are shown as their own line (FR-FIN-030) — a settlement that is short by the fee is not a discrepancy and must not look like one.
- Every unmatched item has one obvious next action, never just a red flag.

## 8.9 Group Roll-Up (Staff Console, multi-school tenants only)

The capability the old single-school architecture could not express. Appears only when the tenant has more than one school (FR-ANL-030).

- Schools as rows, metrics as columns — collections, attendance, pass rate, outstanding actions.
- Ranked comparison with the tenant's own average as the baseline, not an industry benchmark that does not exist.
- Drill from any cell into that school's own dashboard, with the context switcher following the drill.

## 8.10 Impersonation Banner (Platform Console)

TEN-021 and TEN-022 require impersonation to be time-boxed, ticket-linked and visible to the tenant in real time. The interface must make it impossible to forget you are inside a customer's account.

■ SUPPORT SESSION · Sunrise Basic School ■  
Ticket #4821 · expires in 47 min · read-only  
This session is visible in the school's audit log.  
[ End session now ]

Full-width, high-contrast, non-dismissible, present on every screen for the duration. Actions blocked for the platform role (reversal, export, permission change) are visibly disabled with the four-eyes requirement named — a deliberate departure from 34.5's default pattern of hiding unavailable actions; see §14 for why, rather than leaving it unstated here.

## 8.11 Score Entry (Teacher Field App, offline) — fixes a v1.0 gap

FR-UX-020 names attendance and score entry together, with equal force: 'the two highest-frequency, most connectivity-sensitive workflows.' v1.0 gave attendance a full spec and the score-entry equivalent a single line in the inventory table. That asymmetry is corrected here — this screen reuses every mechanism the register established (pre-fetch, IndexedDB queue, SyncLedger) rather than inventing new ones, because a teacher should not have to learn two different offline behaviours in one app.

← Maths · Mid-term · JHS 2A    Grid ▢  
● Working offline · 18 of 32 on this phone

---

Ama Mensah

|   |   |   |   |  |
|---|---|---|---|--|
| - | 8 | 4 | + |  |
|---|---|---|---|--|

out of 100

✓ on this phone

---

Kofi Boateng

|   |     |   |   |  |
|---|-----|---|---|--|
| - | 100 | 0 | + |  |
|---|-----|---|---|--|

out of 100

✓ saved to school

---

18 of 32 scored  
[ Submit when back online ]

- One student per screen section, not a spreadsheet — the Staff Console grid (§8.3) is correct for a desktop mouse and keyboard; a numeric stepper with a large tap target is correct for a thumb. Same underlying data, two different input shapes for two different devices, per §3's design posture.
- The -/+ stepper is the primary input; a tap on the number opens the OS numeric keypad for direct entry, satisfying FR-ASM-030's bounds at input time on both paths.

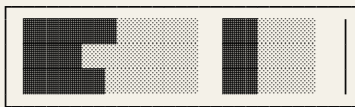


- A 'Grid' toggle in the header switches to a compact all-students-one-screen view for a teacher who prefers scrolling over stepping, once scores have been reviewed once in the detailed view — an efficiency option, not the default.
- Reuses SyncLedger and the identical 'on this phone' / 'saved to school' language from the register (§8.1–8.2), so a teacher learns the offline model once.
- Bounds, missing-score status and the submitted lock (FR-ASM-030, FR-ASM-040) apply identically to this surface and to the Staff Console grid — one validation rule, enforced server-side, expressed in two UIs.

## 8.12 Loading, Empty & Restricted States — fixes a v1.0 gap

34.5 names six required states: loading, success, error, offline, empty, restricted. v1.0 specified and demonstrated success (§8.4, pills), error (§11.1, form errors) and offline (§8.1–8.2) in detail, but named the other three only as component names in the library (§5) with no actual specification. All six are specified here.

### Loading



← skeleton rows, not a spinner  
for anything list-shaped

Skeleton screens matching the target layout's shape for lists and tables (DataTable, dashboards); a centred Spinner only for actions with no predictable shape (e.g. report-card batch generation, which already has its own named progress indicator per NFR-PERF-023). A skeleton must never persist past 5 seconds without becoming an error state — a frozen skeleton is a worse lie than a spinner.

### Empty

No applications yet  
Applications will appear here once a  
parent submits one, or you can add one  
directly.

[ Add application manually ]

Per the writing guidance this document follows: an empty screen is an invitation to act, not an apology. Every EmptyState names what will make the list stop being empty and, where a direct action exists (add a student, create an invoice), offers it in place. A results list empty because nothing has been marked yet reads differently ('Scores haven't been entered for this term') from one empty because a filter excludes everything ('No students match these filters — clear filters').

### Restricted

You don't have access to this record

Health records are restricted to staff  
with the Health Officer role. Ask your  
administrator if you need access.

Reached when a permission-generated menu correctly hid a link, but a direct URL or a stale bookmark reaches the screen anyway — the case 34.5's own principle predicts, since the server, not the menu, is the actual authority. States what is restricted and who to ask, never a bare 403 or a silent redirect to the dashboard, which reads as a bug rather than a boundary.

## 9. Offline & Sync UX

### 9.1 Attendance & Score Sync

Implements FR-UX-020 and FR-UX-030. Scope is deliberate: attendance and score entry are offline-capable. Nothing else is, in v1.

| Concern                           | Behaviour                                                                                                                                                                                 |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre-fetch                         | On any successful connection, the teacher's assigned classes, rosters, active assessments and current term config are cached. Cache is tenant-namespaced (TEN-040) and cleared on logout. |
| Capture                           | Writes go to IndexedDB first, always. The UI never waits on the network to acknowledge a mark or a score.                                                                                 |
| Queue visibility                  | SyncLedger (§8.2) states the count and the state in words at all times on offline-capable screens.                                                                                        |
| Sync                              | Background sync on reconnection; manual retry available. Idempotency keys prevent double-posting on retry (NFR-API-010).                                                                  |
| Conflict — same user, two devices | Last write by device timestamp wins, silently. Stated in the queue detail so it is discoverable.                                                                                          |
| Conflict — different users        | Surfaced for manual reconciliation, never auto-resolved (FR-ATT-011). Both values, both authors, both times, one decision.                                                                |
| Failure                           | Named and actionable: what failed, why, what to do. Never a generic retry toast.                                                                                                          |
| Honesty rule                      | No optimistic UI that implies server persistence. 'Saved on this phone' and 'saved to school' are different sentences and the interface always uses the true one.                         |

### 9.2 PWA Infrastructure — fixes a v1.0 gap

FR-UX-010 requires the application to be installable as a PWA. v1.0 asserted this in the surfaces table and never specified a manifest, an icon set, or a caching strategy — and the companion prototype shipped as a plain static page with neither. Both are corrected as of this revision: the prototype now ships with a real manifest.json and service worker (sw.js), described below rather than only claimed.

| Element                                 | Specification                                                                                                                                                                                                                                                      |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Web app manifest                        | name 'PBSMS', short_name 'PBSMS', display 'standalone', start_url scoped to the Teacher Field App entry route, theme_color --pb-teal-900, background_color --pb-canvas, icons at 192px and 512px (maskable + any purpose).                                         |
| Install prompt                          | Deferred beforeinstallprompt captured and surfaced as an in-app 'Install PBSMS' action after a teacher's second successful register submission, not on first load — install offers are more credible once the app has proven useful once.                          |
| Service worker — app shell              | Cache-first for the shell, JS/CSS bundles and icon set, versioned and invalidated on deploy (a stale shell must never silently outlive a release).                                                                                                                 |
| Service worker — roster/assessment data | Network-first with a cache fallback, refreshed on every successful connection per §9.1's pre-fetch rule — never served stale-only.                                                                                                                                 |
| Service worker — queued writes          | Background Sync API registers pending attendance/score submissions; on 'sync' event, the queue in IndexedDB is flushed through the same idempotent endpoints §9.1 already specifies. The service worker is a delivery mechanism for the queue, not a second queue. |

*Honest limitation carried into this revision: a service worker requires an https or localhost origin to register — it will not activate if the prototype HTML is opened directly as a local file (file:///). The companion manifest.json and sw.js are real, are referenced from the prototype, and will activate when served through any local dev server (e.g. npx serve) or once deployed — but double-clicking the HTML file will not demonstrate installability. This is stated here rather than left for someone to discover by a failed install prompt.*

### 9.3 Data-Saver Mode — fixes a v1.0 gap

FR-UX-040 requires 'a data-saver mode for report viewing.' v1.0 listed pagination and lazy image loading as general low-data practice but never specified this mode as its own concrete behaviour, on Parent View specifically, where it matters most.

- A visible toggle on any report-length screen (report card, statement of account): 'Data saver' — off by default on Wi-Fi-detected connections (via the Network Information API where available), on by default when the connection is detected as 2G/3G or save-data is signalled by the browser.
- With data saver on: photos (student photo, school logo) are replaced by initials/text placeholders; PDF attachments are not auto-fetched, replaced by a named download link with file size shown; charts on analytics screens render as a compact numeric table instead of an SVG chart.
- Data saver never hides data, only defers heavier renderings of the same data — a parent on data saver still sees every score and every cedi, just not the school photo alongside it.
- The toggle's state persists per device, not per session, so a guardian on a fixed, expensive data plan does not re-decide this on every visit.

## 10. Role-Based Views

Chapter 13 defines scope from tenant down to record ownership. The interface renders permission-generated navigation (Chapter 34.1) — but per Chapter 34.5, hiding a control is a usability courtesy and never an authorization mechanism. The server decides; the UI merely declutters.

| Role                    | Lands on                                             | Sees                                                                    | Does not see                                                                      |
|-------------------------|------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Proprietor / Director   | Group roll-up if multi-school, else school dashboard | All schools in tenant, finance summary, academic summary                | Other tenants. Individual health and discipline detail unless separately granted. |
| Headmaster              | Approvals & exceptions dashboard                     | Whole school: academics, attendance, finance summary, approvals         | Platform settings, other schools unless assigned.                                 |
| Accountant / Bursar     | Finance dashboard                                    | Invoices, payments, allocations, reconciliation, statements             | Academic scores, health, discipline.                                              |
| Academic Coordinator    | Results workflow queue                               | Assessment config, results pipeline, curriculum, analytics              | Finance, payroll, health.                                                         |
| Class / Subject Teacher | Teacher Field App home                               | Own assigned classes and subjects only                                  | Other teachers' classes, finance, admin settings.                                 |
| Admissions Officer      | Application queue                                    | Applications, duplicates, conversion                                    | Enrolled-student finance and academic records beyond conversion.                  |
| Parent / Guardian       | Parent home                                          | Linked children only; published results, attendance, invoices, messages | Any other child. Unpublished or draft results.                                    |
| Student                 | Student home                                         | Own published results and timetable                                     | Finance, discipline, other students.                                              |
| Platform Support        | Tenant list                                          | One tenant at a time, under an active audited grant                     | Financial reversal, export, permission change without a second approver.          |

## 11. Accessibility Implementation

Chapter 42 binds the product to WCAG 2.1 Level AA with a CI gate (NFR-ACC-020) and a pre-GA manual audit (NFR-ACC-030). Naming a standard is not implementing one — this section is what conformance actually requires here.

### 11.1 Concrete commitments

| WCAG criterion             | Implementation in PBSMS                                                                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.4.3 Contrast (Minimum)   | All token pairs verified at 4.5:1 body / 3:1 large. Tenant brand colours validated on upload and rejected if they fail (§4.4).                          |
| 1.4.1 Use of Colour        | Every status pairs colour with a glyph and a text label. Attendance states, sync states, result states and payment states all readable in greyscale.    |
| 2.1.1 Keyboard             | Score grid, approval flows, allocation and every dialog fully operable by keyboard. The score grid is designed keyboard-first, not keyboard-compatible. |
| 2.4.7 Focus Visible        | 2px focus ring in --pb-teal-600 with 2px offset, never removed. Focus is trapped in modals and returned on close.                                       |
| 2.5.5 Target Size          | 44×44 CSS px minimum on Teacher Field App and Parent View — the surfaces used on phones one-handed.                                                     |
| 3.3.1 / 3.3.3 Errors       | Errors name the field, state what is wrong and state how to fix it. Form-level summary lists every error with in-page links.                            |
| 4.1.3 Status Messages      | SyncLedger, toasts and validation summaries announce via aria-live so a screen-reader user learns a register synced without moving focus.               |
| 1.3.1 Info & Relationships | Data tables use real th/scope. The score grid uses a grid role with correct row and column semantics, not a div soup.                                   |

### 11.2 Verification, not assertion

Every contrast value in §4.1 must be re-checked against a contrast tool before implementation, and the CI accessibility job must fail the build on new critical violations from the first frontend commit — not added once the component library already exists. Retrofitting accessibility across an existing component set is the same category of expensive mistake as retrofitting multi-tenancy across an existing schema.

Fixed v1.0 gap: NFR-ACC-020 was written into this document's previous revision but did not exist in the code scaffold's actual CI pipeline — a requirement stated in the spec and absent from the artifact meant to enforce it. A pa11y-ci job now exists in `.github/workflows/ci.yml`, running against the built Staff Console and Parent View, blocking merge on new critical violations. It is a starting gate, not the NFR-ACC-030 manual audit — the two remain distinct and both remain required before General Availability.

## 12. Performance Budget

NFR-PERF-020 requires p95 page load  $\leq 2.5s$  on a 3G-equivalent connection. A budget makes that testable per screen rather than aspirational.

| Surface           | JS (gzipped)  | CSS          | Fonts                    | Total first load | Rationale                                                             |
|-------------------|---------------|--------------|--------------------------|------------------|-----------------------------------------------------------------------|
| Parent View       | $\leq 60$ KB  | $\leq 15$ KB | 0 KB (system stack only) | $\leq 150$ KB    | Metered data, arrives from a link, must never feel expensive to open. |
| Teacher Field App | $\leq 140$ KB | $\leq 20$ KB | $\leq 28$ KB             | $\leq 250$ KB    | Installed once, cached thereafter; offline shell justifies more.      |
| Staff Console     | $\leq 250$ KB | $\leq 30$ KB | $\leq 28$ KB             | $\leq 400$ KB    | Desktop, office connection, route-split per module.                   |
| Platform Console  | $\leq 250$ KB | $\leq 30$ KB | $\leq 28$ KB             | $\leq 400$ KB    | Internal use only.                                                    |

- Route-level code splitting per module; no module's code loads for a user whose role cannot reach it.
- Lists paginate by default (FR-UX-040); no unbounded fetch anywhere.
- Images: student photos lazy-loaded, served at display size, never full-resolution originals in a list.
- Budget enforced in CI as a failing check, alongside the accessibility gate.

## 13. Build Order

Sequenced so that the two things that cannot be retrofitted — the offline layer and the accessibility baseline — exist before the component library grows.

| Stage | Build                                                                          | Why here                                                                                              |
|-------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1     | Tokens, type scale, base components, focus and contrast baseline, CI a11y gate | Everything else inherits from this. Cheapest moment to get it right.                                  |
| 2     | App shell, auth, context switcher, permission-generated navigation             | Every screen mounts inside it.                                                                        |
| 3     | Offline/sync layer + SyncLedger, proven with the register                      | Hardest to retrofit. Build it against one real screen before there are twenty.                        |
| 4     | Teacher Field App — register, score entry                                      | Highest-frequency users; earliest real feedback; validates stage 3.                                   |
| 5     | Staff Console — students, academic structure, assessment, results              | The operational core.                                                                                 |
| 6     | Parent View — home, results, invoices, pay                                     | Unlocks the communication value proposition.                                                          |
| 7     | Finance console — invoicing, payments, allocation, reconciliation              | Depends on 5; highest correctness risk, so not first.                                                 |
| 8     | Communication, analytics, supporting operations, compliance screens            | Breadth after depth.                                                                                  |
| 9     | Platform Console                                                               | Internal users tolerate rough edges longest; also gated on Chapter 3 impersonation controls existing. |

## 14. Documented Deviations from the SRS

v1.0 contained one place where the actual design departed from the SRS's literal stated pattern while being presented as compliant. That is a worse failure than a missing feature — a missing feature is visible; a silent deviation is not. This section exists so it cannot happen again unnoticed: any future departure from the SRS's stated text belongs here, named, with the SRS clause, the deviation, and the reasoning, reviewed by whoever holds the Engineering Lead role before it ships.

| SRS clause                                                    | Stated pattern                                                | This design                                                                                                                     | Reasoning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 34.5 — 'The frontend hides unavailable actions for usability' | Actions the current user cannot perform are removed from view | Impersonation-blocked actions in the Platform Console (§8.10) are shown, visibly disabled, with the four-eyes requirement named | 34.5's own purpose is to prevent the client from being mistaken for the authority — the server still decides. For this specific screen, a support engineer who cannot see why a button is missing will assume the product is broken and open a second support ticket about the first one. Naming the constraint is more informative than hiding it, and does not weaken the actual authorization boundary, which remains server-enforced either way. This exception is scoped to the Platform Console only — every other surface follows 34.5's literal pattern of hiding, not disabling. |

*No other deviations are known at this revision. This table is expected to grow as the product is actually built and real constraints surface — an empty deviations table after implementation would be a sign this section stopped being used, not a sign nothing was ever traded off.*



## 15. Open Questions & Honest Limitations

This specification is a proposal by one designer working from a document, not from users. The following are genuinely unresolved and should not be treated as settled by the confidence of the prose above.

- No user research exists. Nothing here has been shown to a Ghanaian teacher, headmaster, bursar or parent. The register design in §8.1 is the highest-conviction screen and also the one most likely to be wrong in a way only a real teacher would catch. Test it before building stage 4.
- The four-surface split (§3) is the load-bearing structural bet. If it is wrong, most of this document changes. It should be reviewed by whoever holds the Engineering Lead role before stage 1 begins.
- Colour and type are proposals with reasoning, not brand decisions. If PBSMS acquires a brand identity later, §4.1 and §4.2 are the parts to replace — the rest of the document survives that change.
- Contrast ratios in §4.1 are calculated, not tool-verified. Verify before implementation (§11.2).
- The performance budgets in §12 are set from typical figures for this class of application, not from measurement of a PBSMS build that does not yet exist. Re-baseline once stage 4 ships, in the same spirit as NFR-PERF-090.
- Timetable generation is specified as conflict detection only, consistent with SRS Chapter 17 — automatic generation remains unspecified there and is deliberately not invented here.
- This document has now been audited once against the SRS's literal text (v1.0 → v1.1, front matter) and found compliant on that pass. A second audit by someone other than the original author would very likely find a different set of gaps than the first one did — the same blind spots that produced a gap tend to also miss it on self-review. Treat v1.1 as more accurate than v1.0, not as finished.

— End of Document —